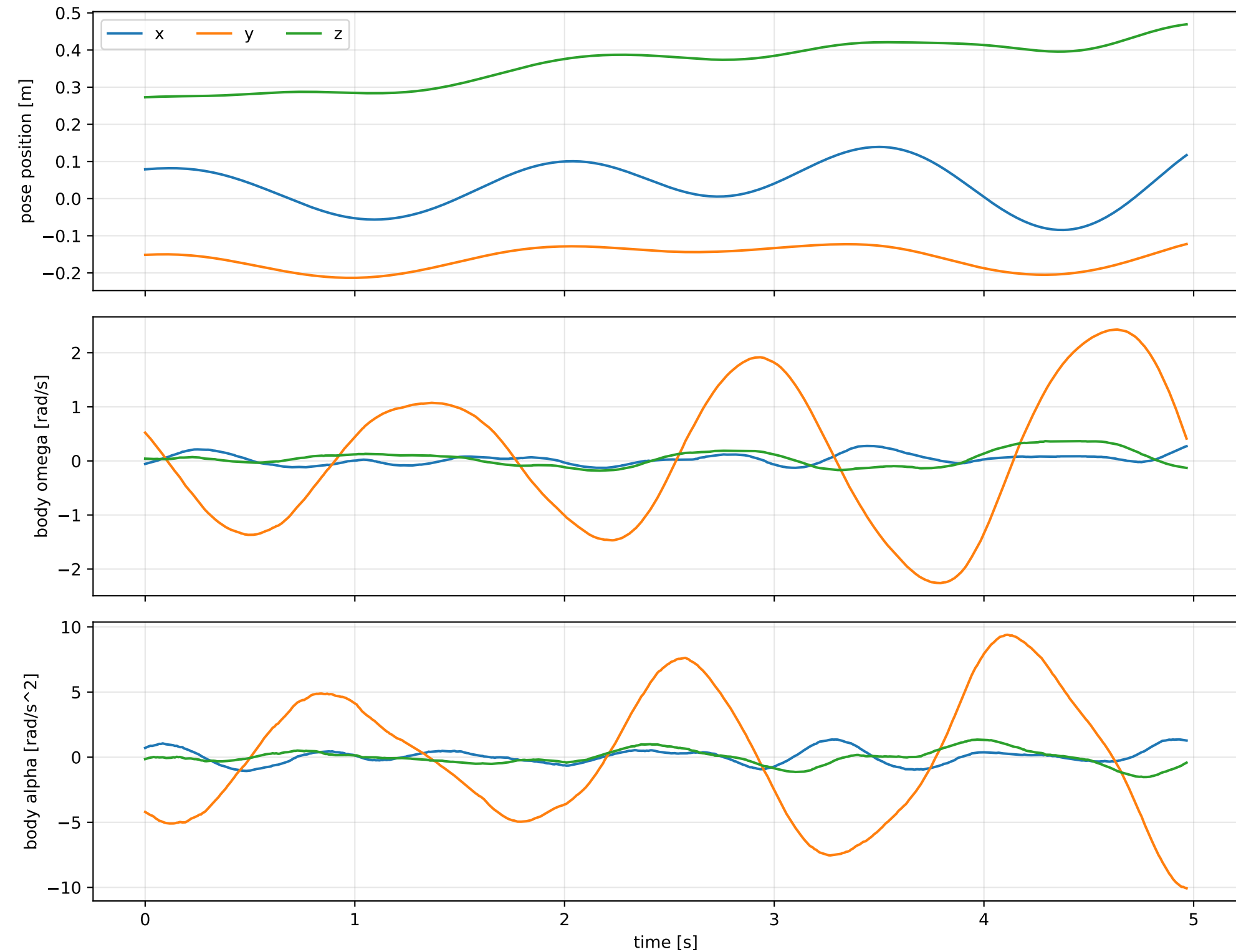
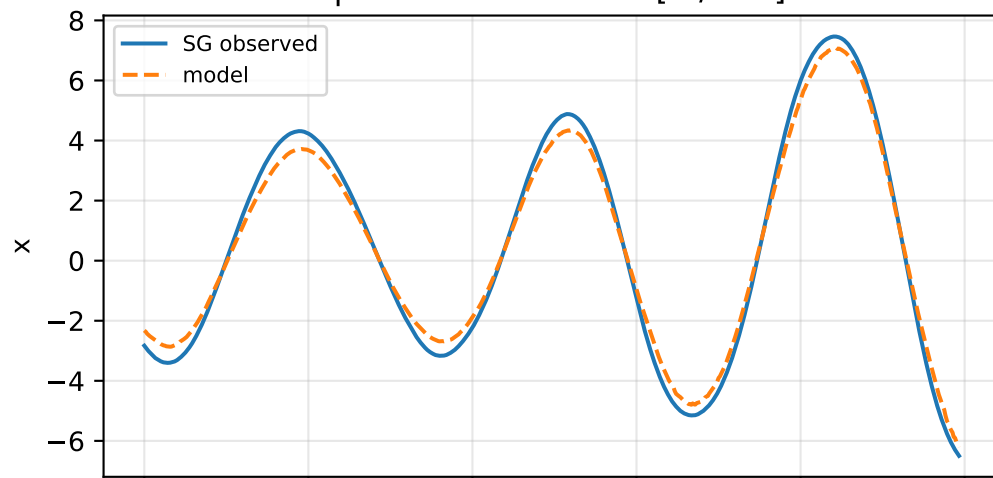
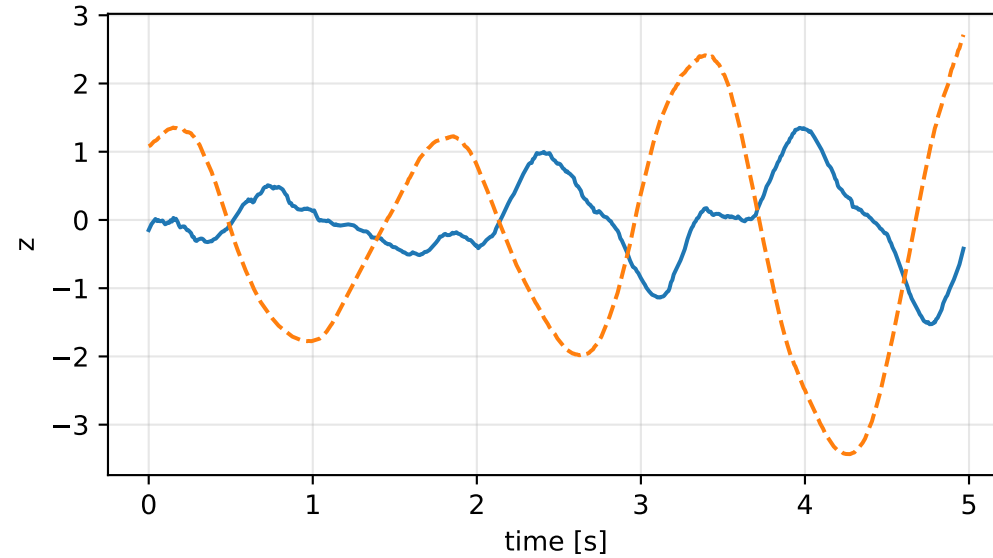
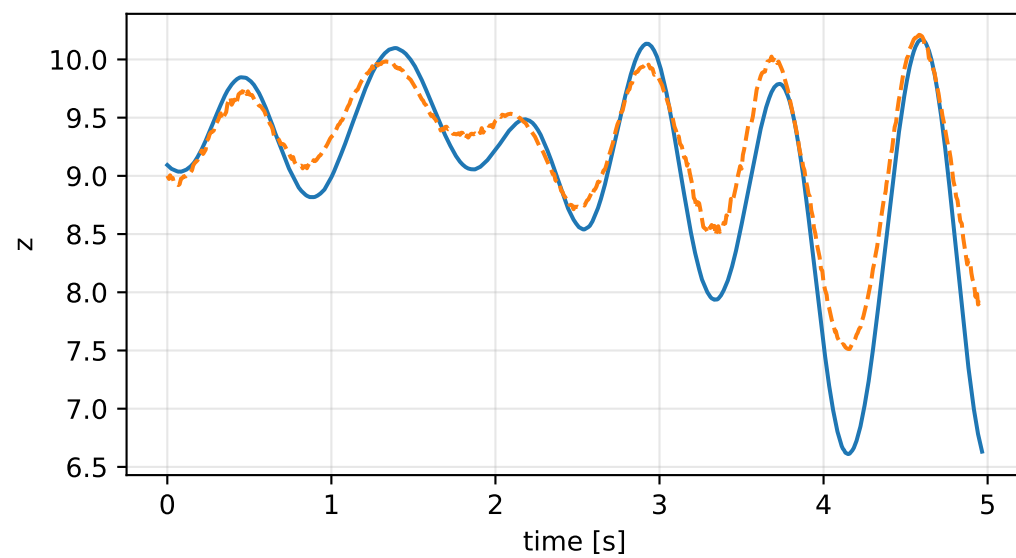
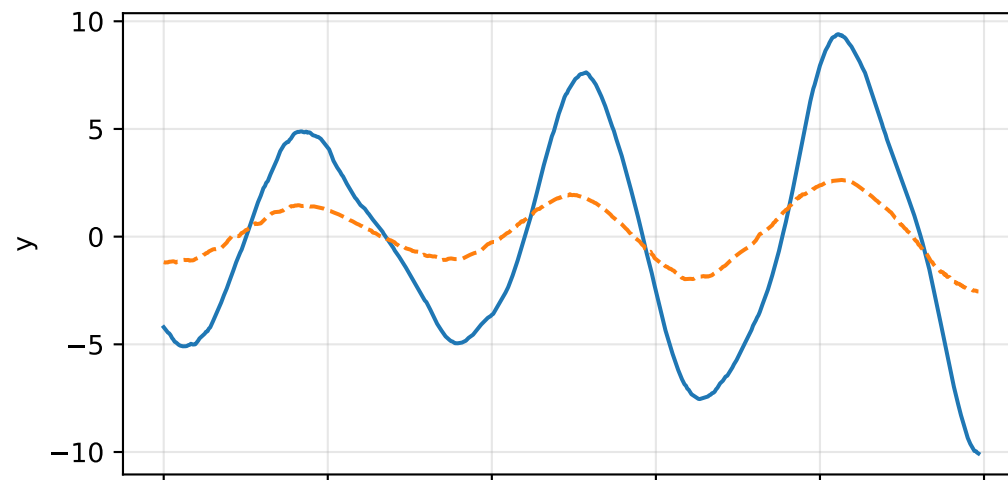
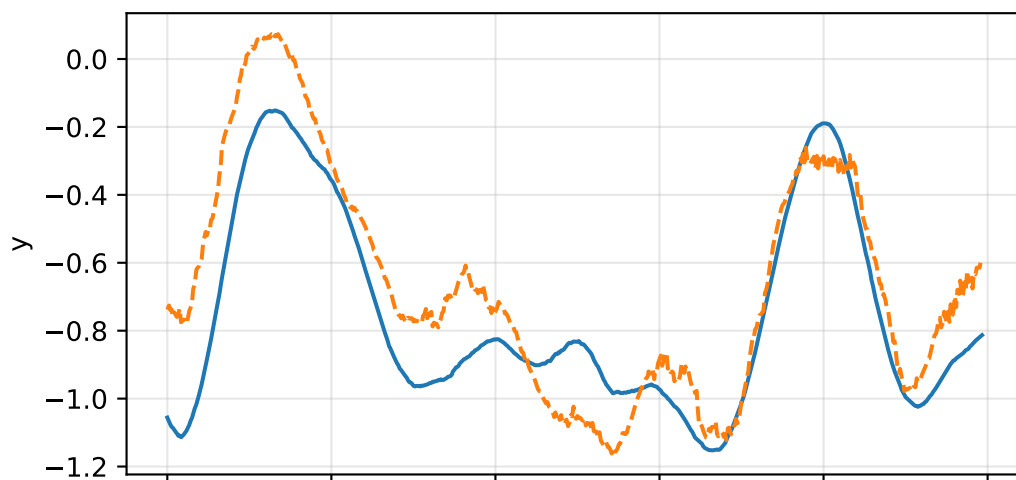
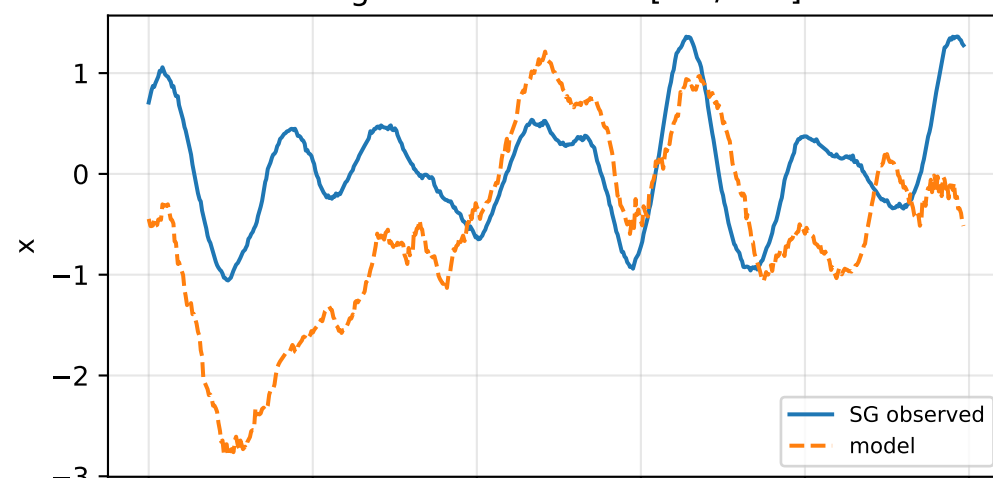


cov_full_no_position_rotation_cross: SG trajectory and derived motion

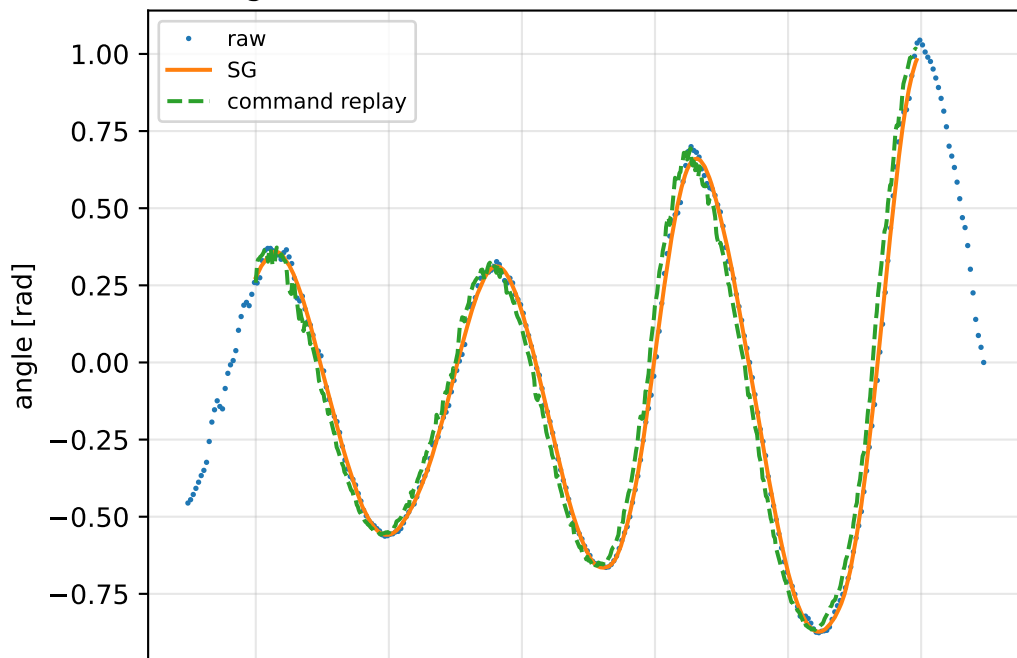


cov_full_no_position_rotation_cross: acceleration residual objective

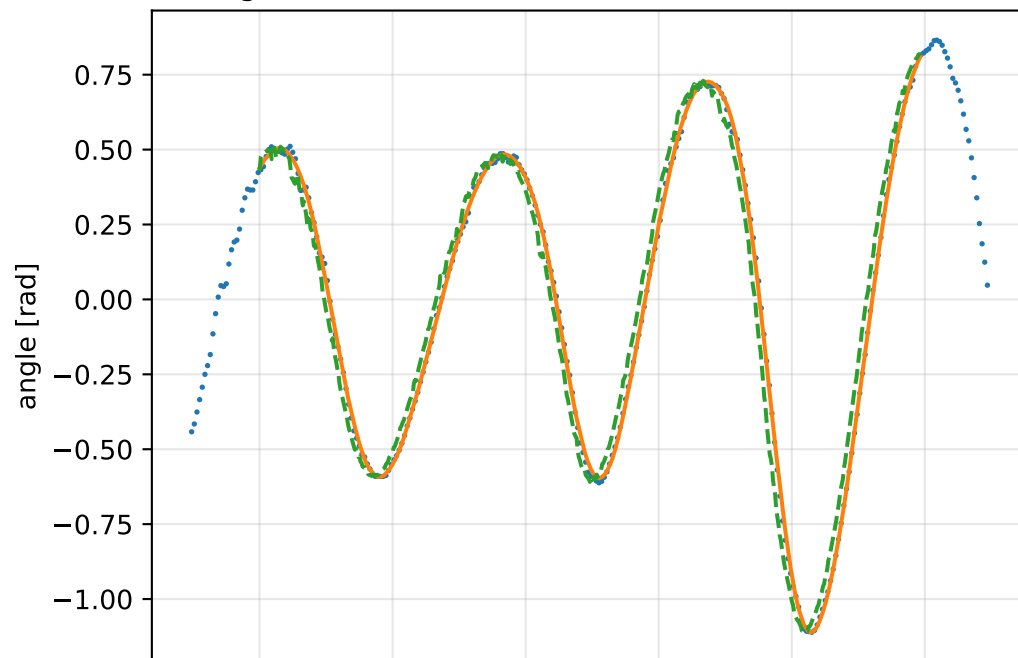
specific acceleration [m/s²]angular acceleration [rad/s²]

cov_full_no_position_rotation_cross: gimbal measurement audit (objective=measured_sg)

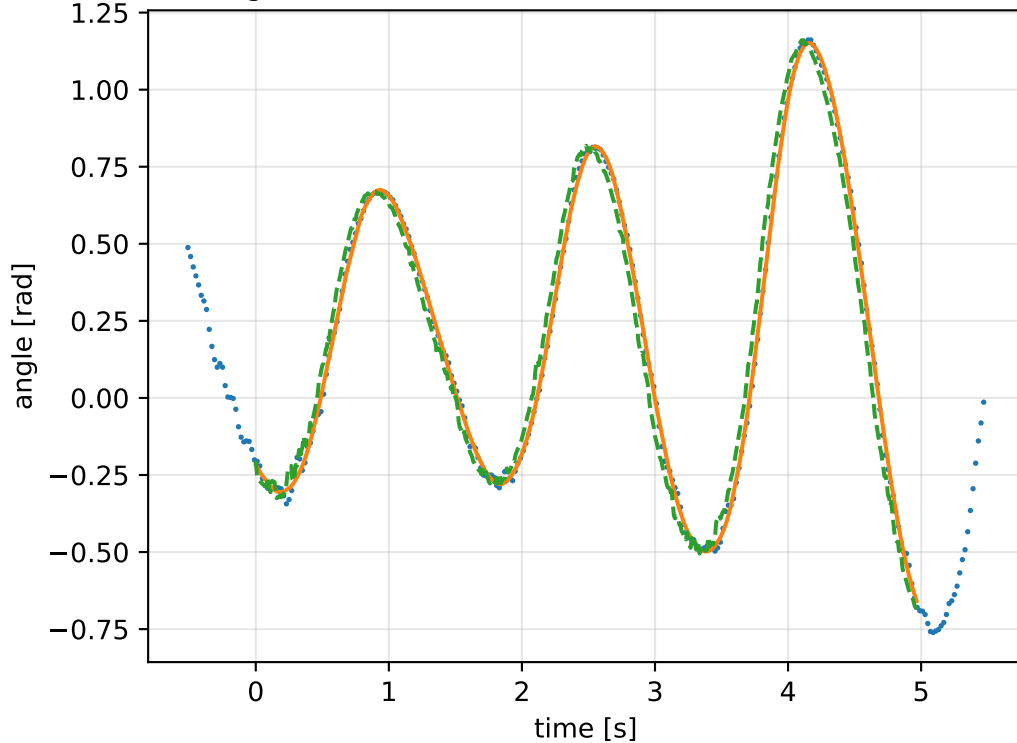
gimbal 1: RMSE=0.0759, max=0.2095 rad



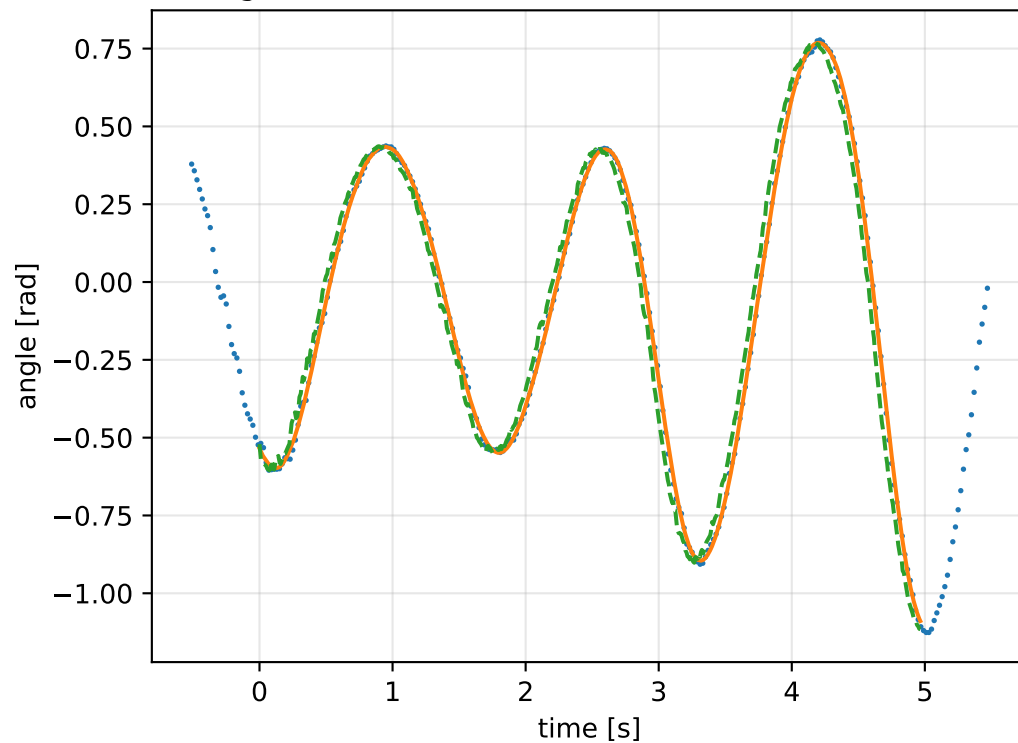
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

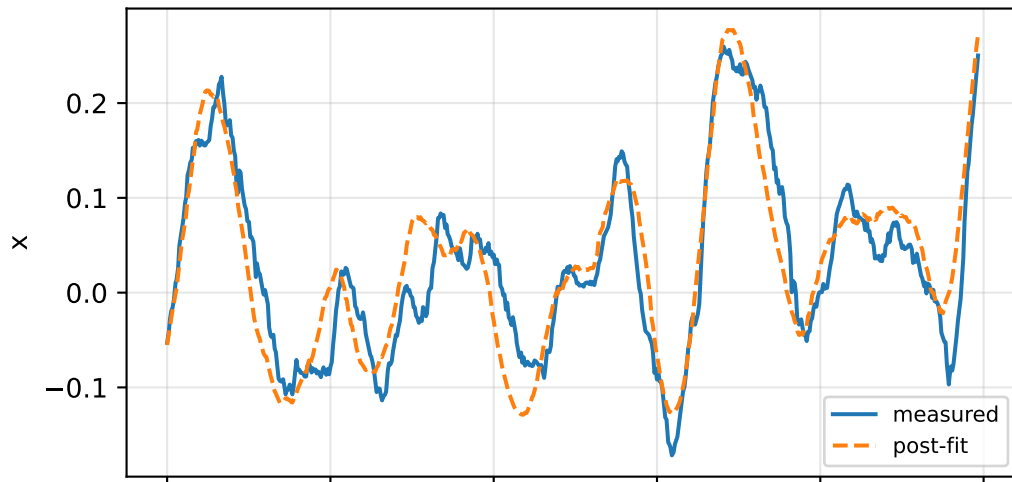


gimbal 4: RMSE=0.07131, max=0.1737 rad

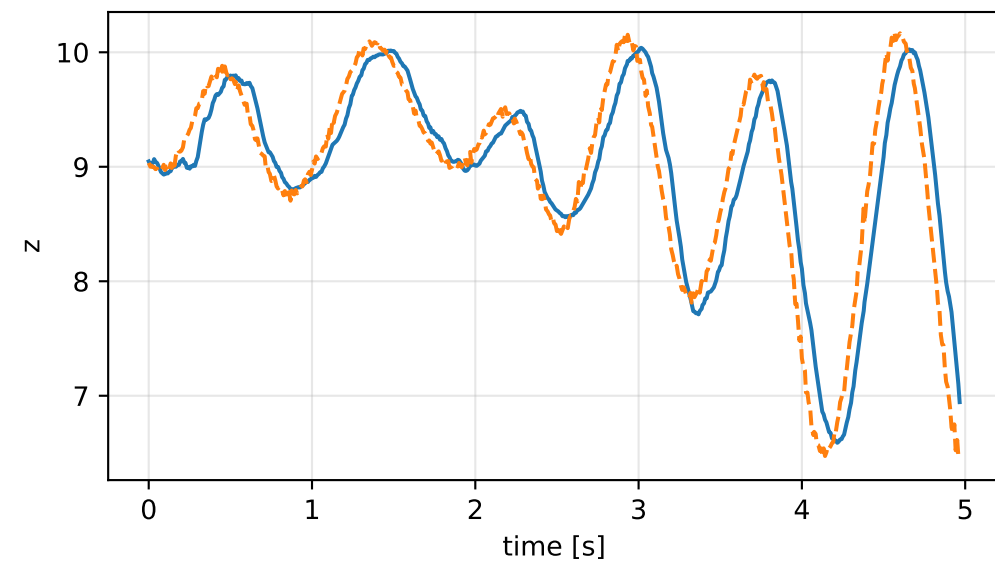
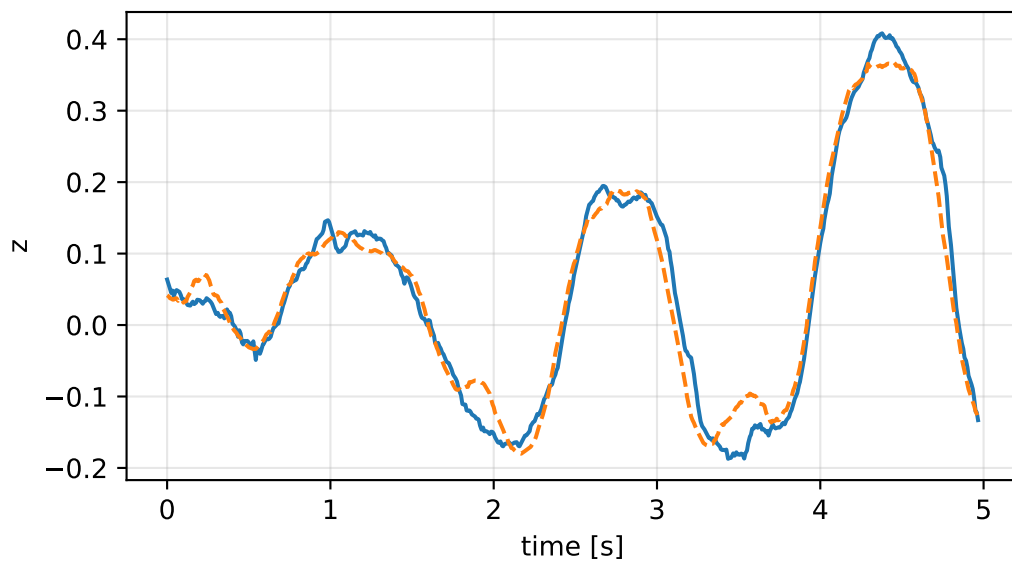
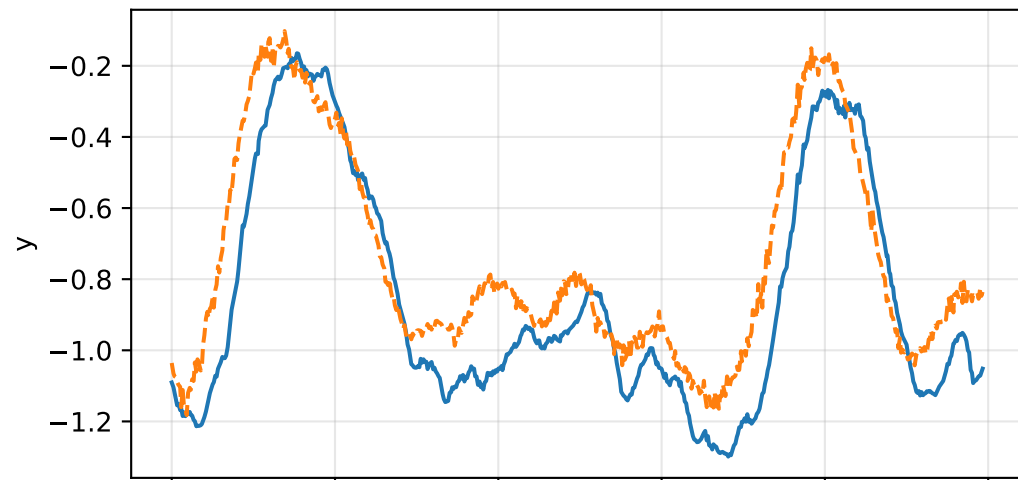
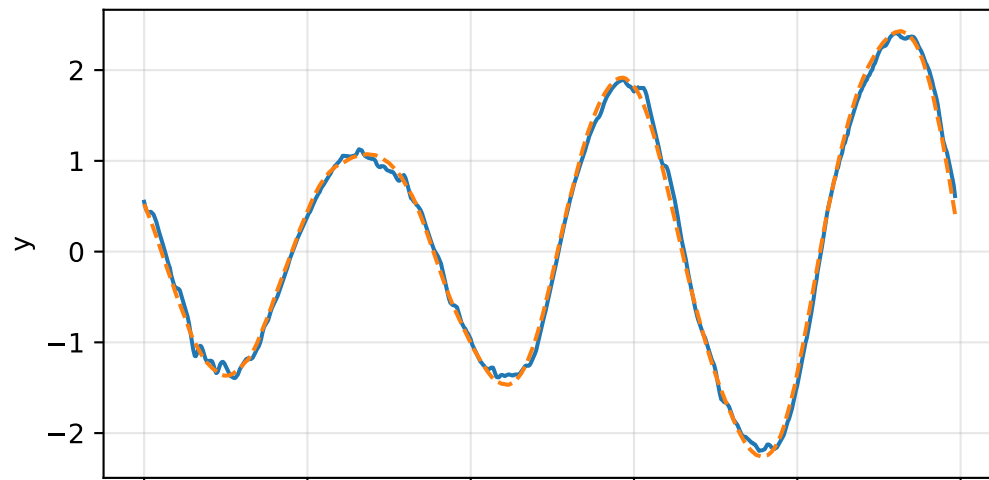
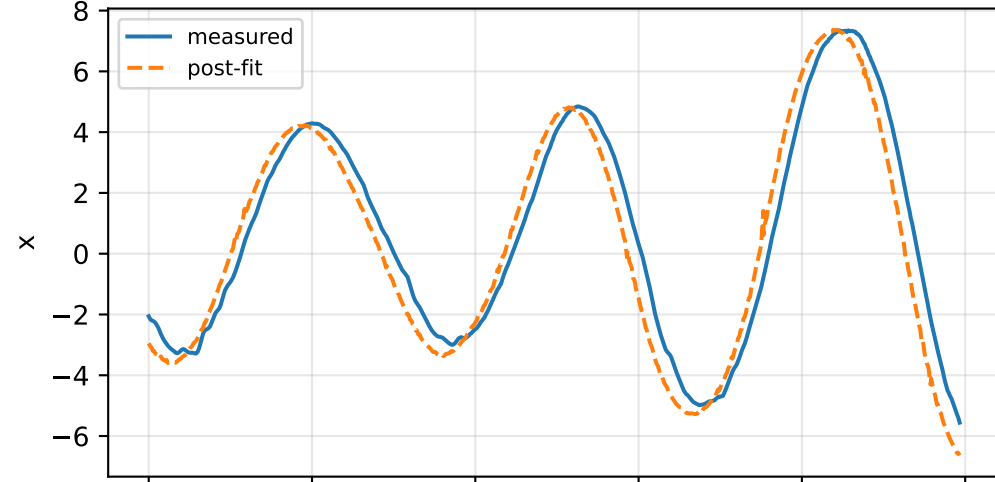


cov_full_no_position_rotation_cross: IMU comparison (diagnostic only)

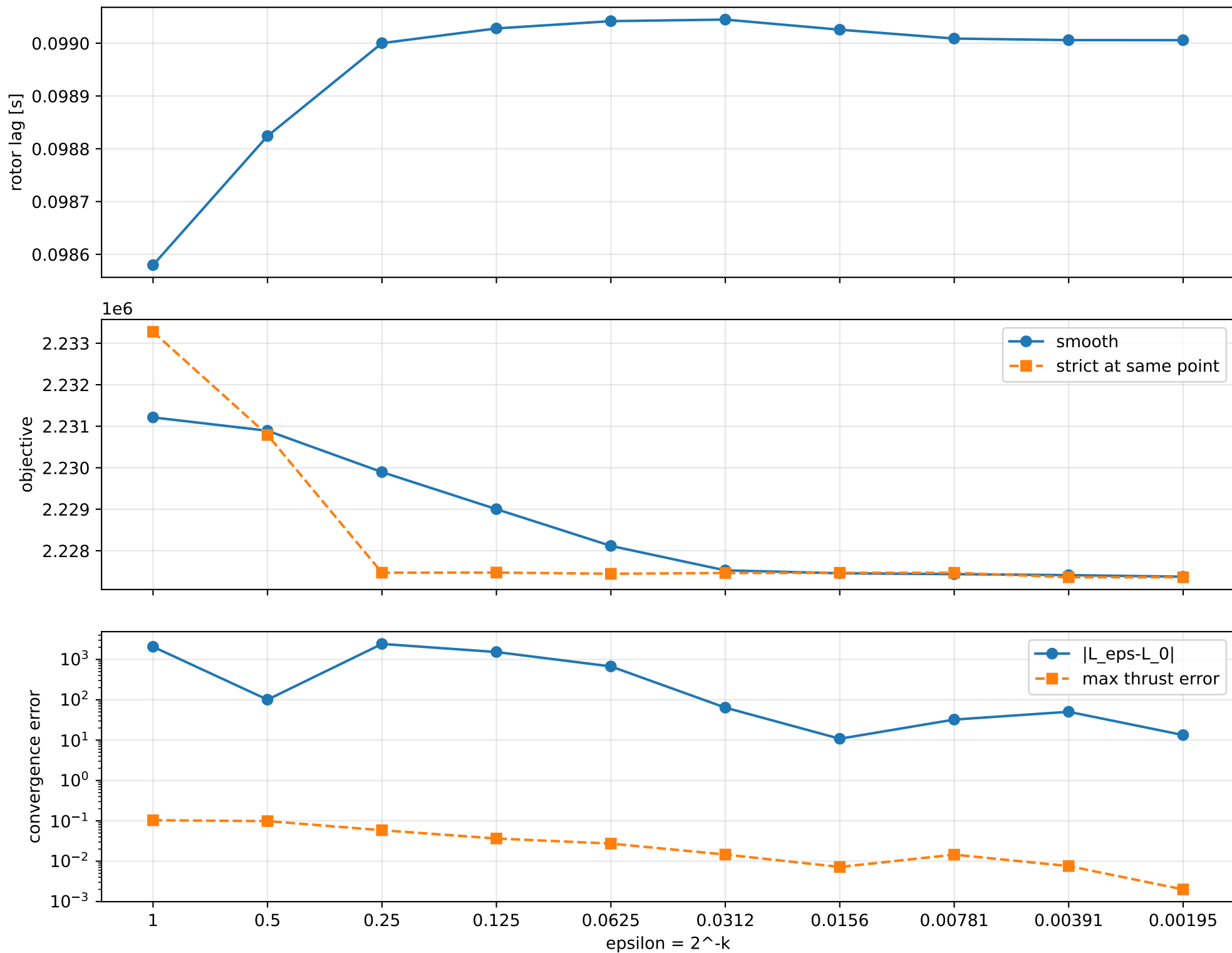
gyro [rad/s]



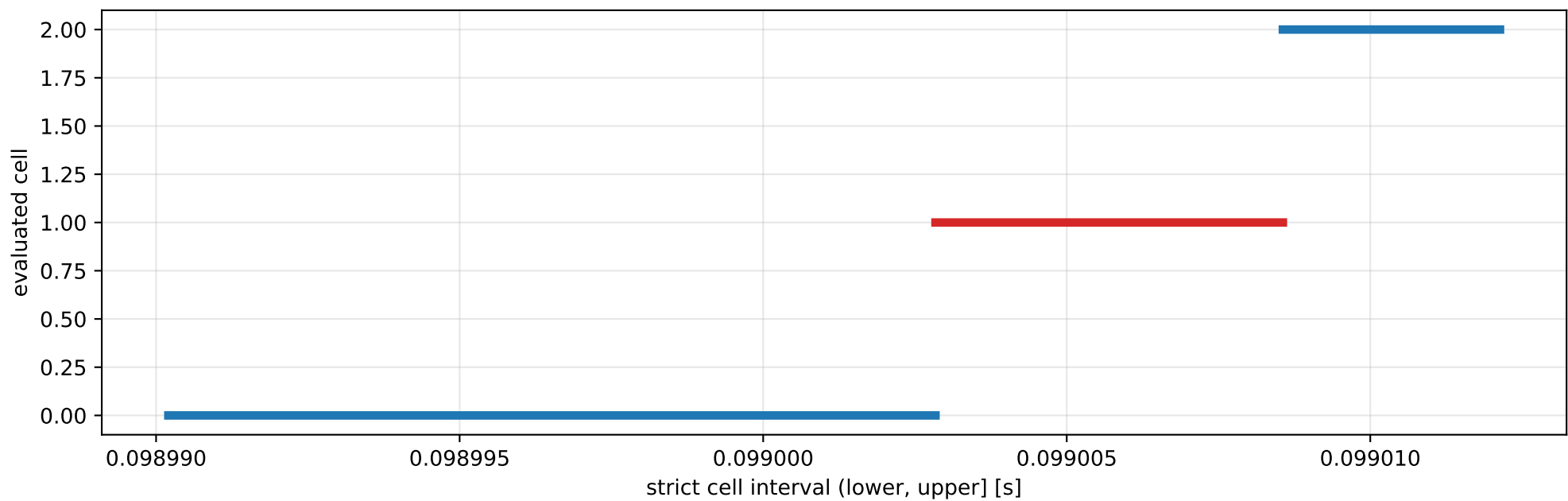
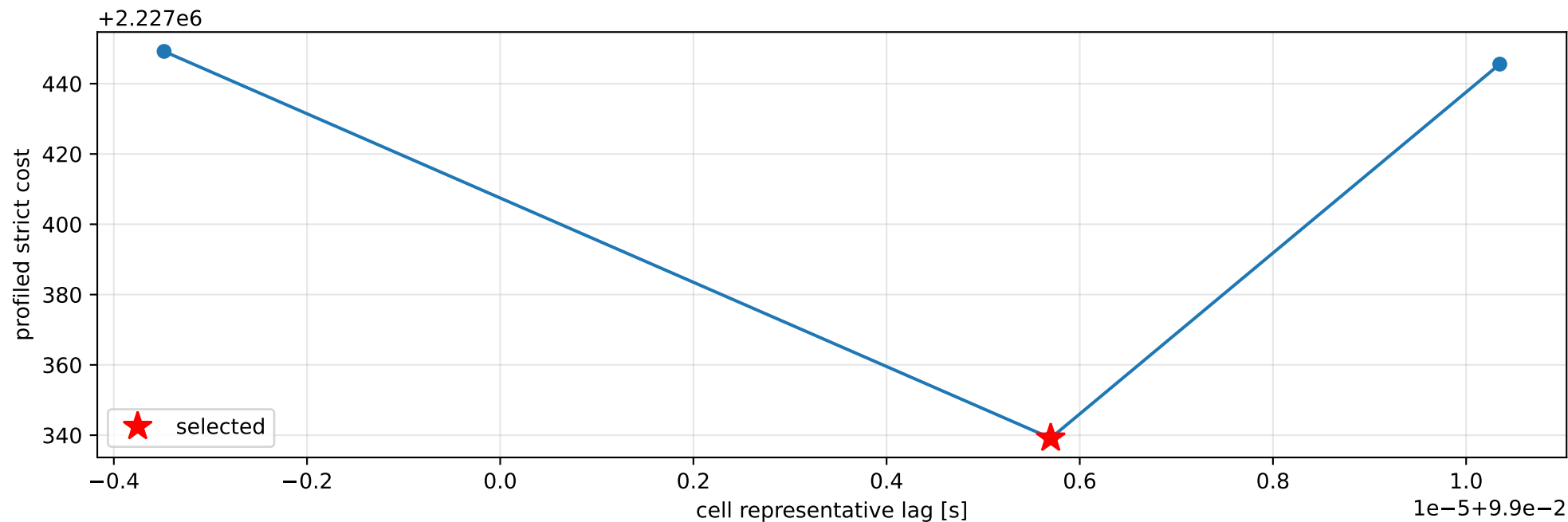
specific force [m/s^2]



cov_full_no_position_rotation_cross: smooth-to-strict continuation



cov_full_no_position_rotation_cross: exact strict-ZOH lag cells



selected (0.0990028381, 0.0990085602] s; final smooth 0.0990057454 s; data boundary=False

cov_full_no_position_rotation_cross: parameter estimate

Case: cov_full_no_position_rotation_cross
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 5.71057946

inertia [kg m²]:

```
[[ 0.12243455 -0.09483993 -0.09209964]
 [-0.09483993  1.81867268 -0.03479084]
 [-0.09209964 -0.03479084  1.91447841]]
```

CoG body [m]: [0.00877564 0.16534405 -0.05314898]

force effectiveness: [0.5856623 0.67632996 3.66977362 3.41405863]

scale-free J/m [m²]:

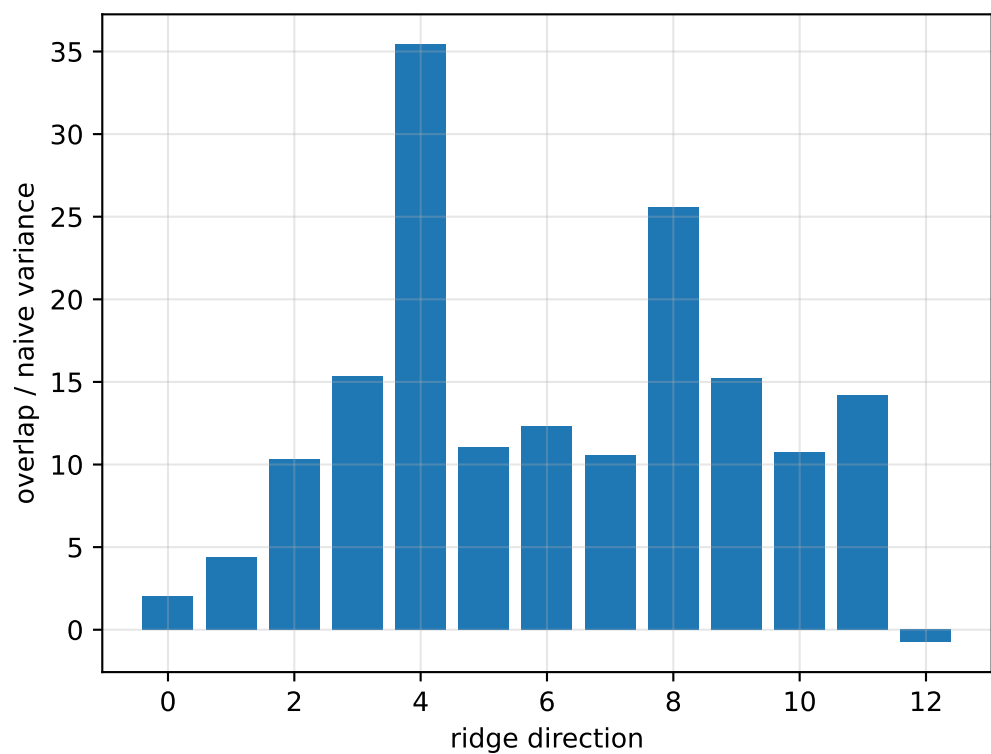
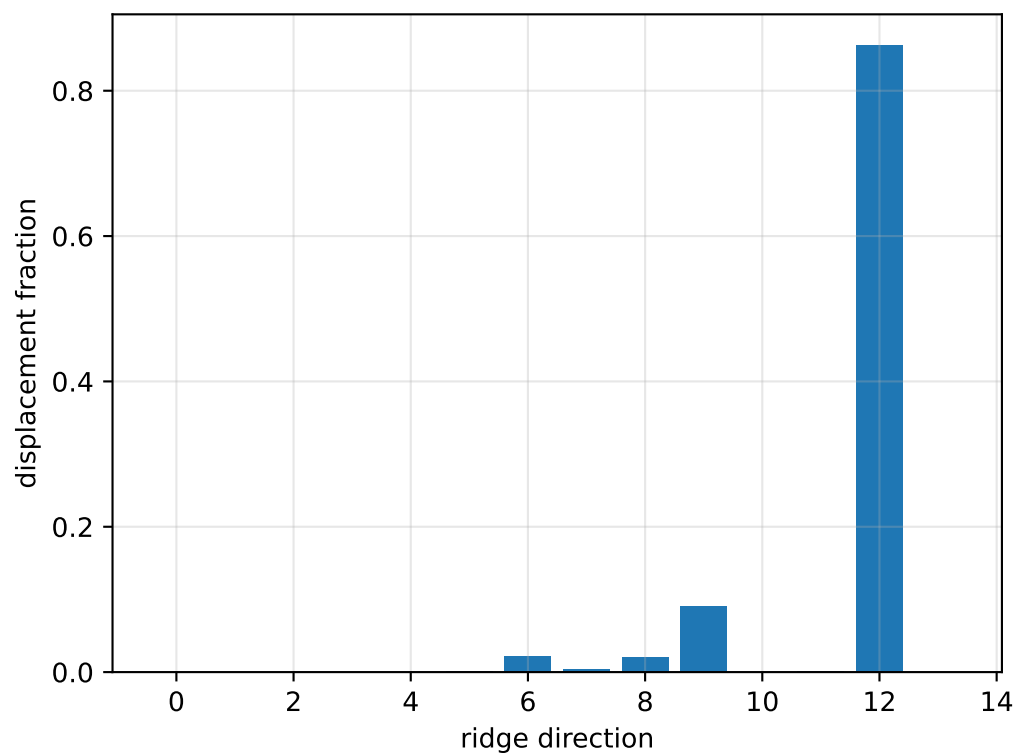
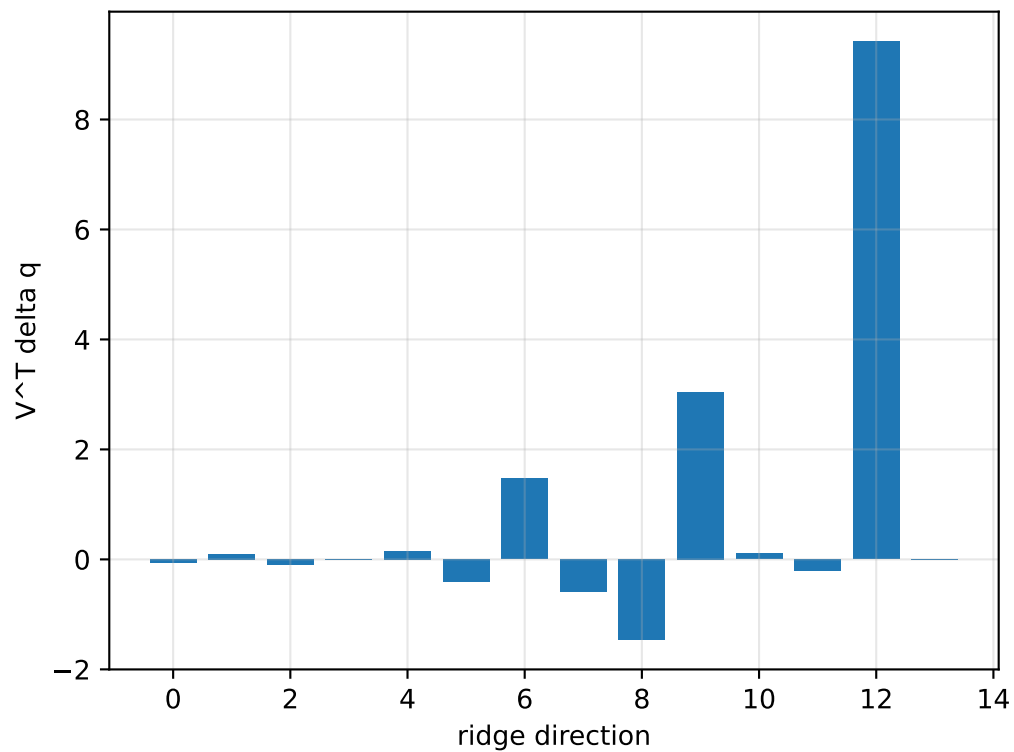
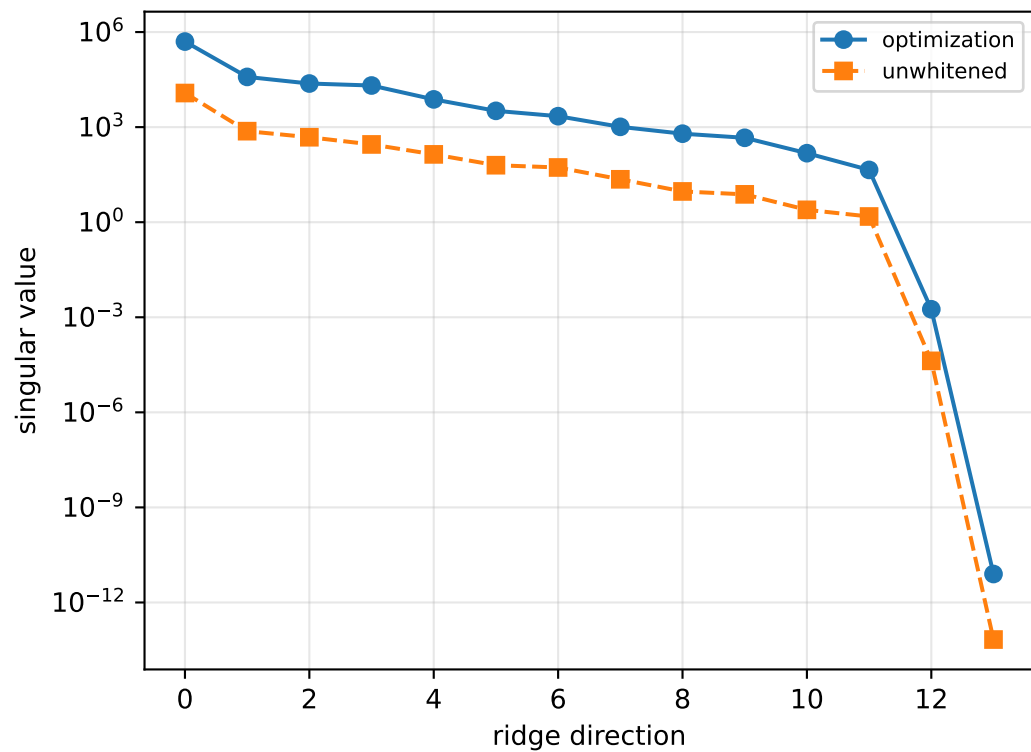
```
[[ 0.02143995 -0.01660776 -0.0161279 ]
 [-0.01660776  0.31847428 -0.00609235]
 [-0.0161279  -0.00609235  0.33525116]]
```

scale-free f/m: [0.10255742 0.11843456 0.64262719 0.59784802]

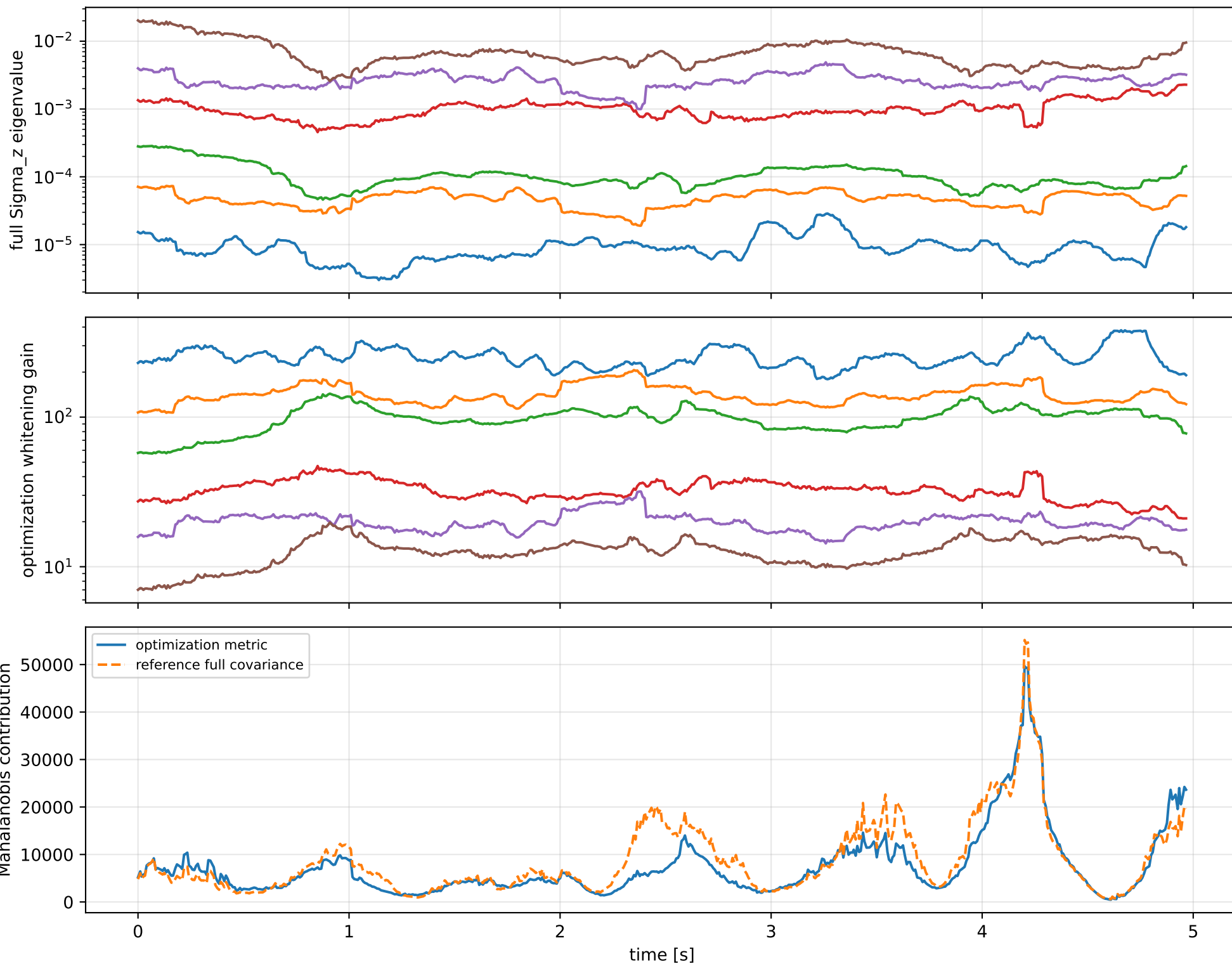
rotor lag cell: (0.0990028381, 0.0990085602] s

representative: 0.0990056992 s

cov_full_no_position_rotation_cross: ridge spectrum (rank 13, nullity 1, ||v||=1.22e-11)



cov_full_no_position_rotation_cross: optimization vs reference covariance



cov_full_no_position_rotation_cross: wrench / closure (force max $2.09\text{e-}14$, torque max $5.33\text{e-}15$)

